



Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE
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Home Assignment 3

- 1) Construct 95% CI for a survey proportion using given data.
- 2) Perform hypothesis testing for population mean & interpret p-value.
- 3) Conduct one-way ANOVA on three groups (sales/trials/samples) using Python.

1: CONSTRUCTION OF 95% CONFIDENCE INTERVAL FOR PROPORTION

Objective

To estimate the range within which the true population proportion lies with 95% confidence.

Given Data

- Total observations (n) = 30
- Number of successes (placed students) = 21

Step 1: Calculate Sample Proportion $\hat{p} = x / n = 21 / 30 = 0.7$

Step 2: Formula for Confidence Interval

$$CI = \hat{p} \pm Z * \sqrt{(\hat{p} (1 - \hat{p}) / n)}$$

Where:

- $Z = 1.96$ (for 95% confidence level)
- $\hat{p} = 0.7$
- $n = 30$

Step 3: Calculate Standard Error

$$SE = \sqrt{(0.7 \times 0.3 / 30)} \quad SE = \sqrt{(0.21 / 30)} \quad SE = \sqrt{0.007} = 0.0837$$

Step 4: Calculate Margin of Error

$$\text{Margin} = 1.96 \times 0.0837 = 0.164$$

Step 5: Compute Confidence Interval

$$\text{Lower Limit} = 0.7 - 0.164 = 0.536 \quad \text{Upper Limit} = 0.7 + 0.164 = 0.864 \quad \text{Final}$$

Answer

$$95\% \text{ Confidence Interval} = (0.536, 0.864)$$

Interpretation

We are 95% confident that the true proportion of placed students lies between 53.6% and 86.4%. This indicates a relatively high placement rate.

2: HYPOTHESIS TESTING FOR POPULATION MEAN

Objective

To test whether the sample data significantly differs from a given population mean.

Given Data

- Sample size (n) = 14
- Sample mean ($\bar{x}$) = 499.93
- Sample standard deviation (s) = 3.75
- Population mean (μ_0) = 500

Step 1: Formulate Hypotheses

Null Hypothesis (H_0): $\mu = 500$ Alternative Hypothesis (H_1): $\mu \neq 500$ This is a two-tailed test.

Step 2: Choose Test

Since population standard deviation is unknown, we use a t-test.

Step 3: Test Statistic Formula $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$ Step 4:

Calculation $t = (499.93 - 500) / (3.75 / \sqrt{14})$ $t \approx -0.071$

Step 5: Determine p-value

Using t-distribution with degrees of freedom ($df = 13$):

p-value ≈ 0.944

Step 6: Decision Rule

- If p-value $< 0.05 \rightarrow$ Reject H_0
- If p-value $> 0.05 \rightarrow$ Fail to reject H_0 Final Decision p-value = $0.944 > 0.05$

Therefore, we fail to reject the null hypothesis.

Interpretation

There is no statistically significant difference between the sample mean and the population mean of 500. The observed difference is likely due to random variation.

3: ONE-WAY ANOVA (ANALYSIS OF VARIANCE)

Objective

To determine whether there is a significant difference between the means of three groups.

Given Data

Group 1 (Sales): 120, 135, 150, 145, 160, 155, 140

Group 2 (Trials): 130, 142, 138, 145, 150, 148, 152

Group 3 (Samples): 160, 170, 165, 180, 175, 172, 168

Step 1: Hypotheses

$H_0: \mu_1 = \mu_2 = \mu_3$ (All means are equal) H_1 : At least one mean is different

Step 2: Method

One-Way ANOVA compares:

- Between-group variation
- Within-group variation

```
from scipy import stats

g1 = [120,135,150,145,160,155,140]
g2 = [130,142,138,145,150,148,152]
g3 = [160,170,165,180,175,172,168]

f_stat, p_value = stats.f_oneway(g1, g2, g3)

print("F-statistic:", f_stat)
print("p-value:", p_value)
```

... F-statistic: 17.302831142568248
p-value: 6.429247027937587e-05

Step 4: Output

- F-statistic ≈ 17.30
- p-value ≈ 0.000064

Step 5: Decision Rule

Since p-value < 0.05 , we reject the null hypothesis.

Interpretation

There is a statistically significant difference among the three groups (Sales, Trials, Samples). This means that at least one group has a different mean performance.

OVERALL CONCLUSION

1. The confidence interval shows that the placement proportion is relatively high.
2. Hypothesis testing indicates no significant difference between the sample and population mean.
3. ANOVA results confirm that there is a significant difference among group performances.

These statistical methods provide meaningful insights into the dataset and help in decision-making.